STANDATA variance 23-BCV-011R1

Building

Polyethylene film for greenhouse enclosures

Date Issued: October 2026

Purpose

To allow the use of products specifically designed for commercial greenhouse facilities. This variance establishes conditions under which polyethylene film may be used in greenhouses occupied by greenhouse staff only, and under certain circumstances, in greenhouses that allow public access.

Discussion

Flame-resistant fabrics conforming to CAN/ULC-S109, “Standard Method for Flame Tests of Flame-Resistant Fabrics and Films,” are intended for use in tents and air-supported structures. Use of such materials for commercial greenhouses may be detrimental to plant growth due to their ability to reduce the transmission of ultraviolet light required for healthy plant development.

Several types of polyethylene films specifically manufactured for greenhouse applications are available in the marketplace. These products incorporate additives designed to optimize plant growth and environmental performance but generally do not conform to the requirements of CAN/ULC-S109.

Code References

Article 3.1.18.5.

3.1.18.5. Flame Resistance

1) Every tent and *air-supported structure* and all tarpaulins and decorative materials used in connection with these structures shall conform to CAN/ULC-S109, “Standard Method for Flame Tests of Flame-Resistant Fabrics and Films.”

Application

This variance applies to commercial greenhouse facilities that fall under the application of the National Building Code – 2023 Alberta Edition (NBC(AE)).

Variance

This variance provides approximately equivalent or greater safety performance with respect to persons and property as provided for by the *Safety Codes Act* and the NBC(AE).

Polyethylene film used to enclose commercial greenhouse facilities is not required to conform to CAN/ULC-S109, “Standard Method for Flame Tests of Flame-Resistant Fabrics and Films,” under the conditions of the following two options:

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Option 1 – Greenhouses Not Open to the Public

1. The greenhouse is used only for the growing of plants (flowers, vegetables, shrubs, trees, etc.);
2. Members of the public do not have access to the greenhouse; and
3. The polyethylene film is not required to conform to CAN/ULC-S109, "Standard Method for Flame Tests of Flame-Resistant Fabrics," provided that it has a flame spread rating not exceeding 25.

Option 2 – Greenhouses with Public Access

1. The greenhouse is one storey in building height and primarily used for the growing, display, and retail sale of plants (flowers, vegetables, shrubs, trees, and similar horticultural products);
2. The building is classified as a Group F, Division 3 major occupancy (low-hazard industrial occupancy);
3. Public access is limited to retail activities directly related to the sale of plants and horticultural products;
4. The polyethylene film is not required to conform to CAN/ULC-S109, "Standard Method for Flame Tests of Flame-Resistant Fabrics," if it has a flame spread rating not exceeding 25;
5. Public access aisles have a clear width of not less than 1.1 m and be arranged to provide access to exits in at least two directions of travel, with no dead-end aisles;
6. Exit doors serving publicly accessible areas are increased in width by not less than 40% above the minimum width required by the NBC(AE) but not less than 1190 mm;
7. Occupant load is calculated using 4.6 m² per person for areas accessible to the public;
8. Every exit door is provided with exit signage in accordance with the NBC(AE);
9. Spatial separation and exposure protection comply with the NBC(AE). Where more than one greenhouse is located on the same property, the spatial separation between greenhouses may be determined in accordance with the provisions applicable to tents and air-supported structures (Subsection 3.1.18.); and
10. A Fire Safety Plan must be prepared, implemented, and maintained in accordance with the National Fire Code – 2023 Alberta Edition. The Fire Safety Plan must address emergency procedures, evacuation procedures, staff responsibilities, fire protection and life safety systems, maintenance requirements, and emergency contact information. Where required by the building or fire authority having jurisdiction, the Fire Safety Plan must be submitted for review and approval prior to occupancy.

For greenhouses that do not comply with either Option 1 or Option 2, the enclosing materials must conform to CAN/ULC-S109, "Standard Method for Flame Tests of Flame-Resistant Fabrics and Films," as required by Article 3.1.18.5.

This VARIANCE is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code – 2023 Alberta Edition

Issued by the Provincial Building Administrator

[Original signed]

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